

Quiz 5

A tornado is a storm with very strong winds moving in a circular pattern. These storms are most common in the middle of the US, but they occasionally strike Maryland. Historical records from the National Oceanic and Atmospheric Administration show that Maryland had 47 “significant” tornadoes in the 75 years between 1950 and 2024.

- a. Based on the statistic that I provided (47 significant tornadoes in 75 years), what distribution would you use to model the number N of significant tornadoes that strike Maryland each year?

$N \sim \text{Poisson}(47/75)$. We choose $\lambda = 47/75$, since $E[N] = \lambda$. We match the expected value of the annual count of tornadoes with the long-run average number of significant tornadoes per year.

- b. According to your model, what is the probability of having at least one significant tornado strike Maryland in any given year?

We use the Poisson distribution: $P(N \geq 1) = 1 - P(N = 0) = 1 - e^{-47/75} \approx 0.4656$.

- c. It is also interesting to look at *gaps* between years with at least one significant tornado. The gap between such years could be 1, 2, ... years: if there are two years in a row with significant tornadoes (as in 2001 and 2002, for example), then the gap is 1 year. If the significant tornadoes skip a year (as in 2002 and 2004, for example, since there was no significant tornado in 2003), then the gap is 2 years. What distribution would you use to model the gaps? Explain your choice briefly, by connecting gaps between tornado years with the “story” of the distribution that you choose.

Distribution for gaps between years with at least one significant tornado:

$X \sim \text{geometric}(0.4656)$

Brief explanation of how tornado-year gaps match the story of this distribution:

For the geometric distribution, we imagine a sequence of independent trials with probability of success p . The geometric distribution counts the number of trials until the first success. After a tornado year, we count years until the next tornado year. The probability of “success” (a significant tornado) each year is 0.4656. We assume that the events of having significant

tornados in various years are independent. This seems reasonable, as long as there are no substantial trends in weather over the period we are considering (probably just a few years).

- d. There was a significant tornado in Maryland in 2024. According to your model, what is the probability of at least one significant tornado by 2027, i.e. what is $P(X \leq 3)$?

Let p be 0.4656 and q be $1 - 0.4656$. According to the formula for geometric probabilities, we have the following:

$$P(X \leq 3) = P(X=1) + P(X=2) + P(X=3) = p + qp + q^2 p \approx 0.8474$$